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# WISER Model TC2-K
# Wireless Data Sensor
# Temperature

## WIRELESS WITH AN EDGE




The WISER Model TC2-K temperature measurement and monitoring system was designed specifically to tackle the challenges of continuous brake rotor temperature measurements. It is small, light, power efficient, easy-to-use and non-invasive. The wireless system has on-board channel amplification and on-board temperature referencing. The system uses 2 K-type thermocouples and offers highly accurate data over a wide temperature range.

The WISER Model TC2-K is comprised of three subsystems: a remote unit, a base unit, and a software interface. The remote unit connects to the thermocouples and is packaged in the standard electronics enclosure shown above, combined with a custom designed attachment base to mount to any application. The base unit plugs directly into a PC USB port for use with the software interface, or into a power supply and data acquisition system via the two analog outputs. The WISER Data Viewer software is used for system configuration and calibration, live monitoring, and data logging.

### BENEFITS

- Small, wireless system enables measurement in places never before accessed
- Long battery life for uninterrupted testing
- High accuracy enables development work on a wide range of applications
- Non-invasive system can be removed, and does not alter the unit under test

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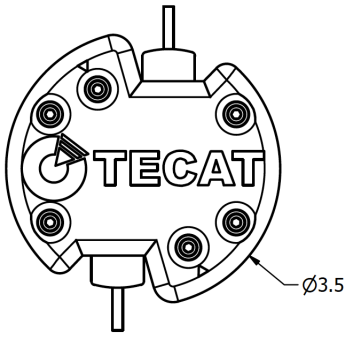
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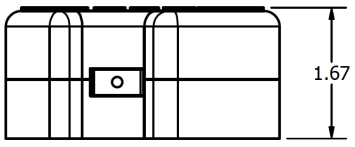
# WISER Model TC2-K Wireless Data Sensor Temperature

## SPECIFICATIONS

| | |
|---|---|
| **Power** | |
| Power Supply Remote Unit | 3V DC to 6V DC, absolute maximum of 6V DC |
| Power Supply Base Unit | 5V DC unregulated |
| Lithium Battery | Standard 3.7V Single Cell LiSOCl2 (550mAh)<br>Option 1: 3.7V Single Cell Li-Poly (400mAh)<br>Option 2: 3.7V Single Cell Li-Poly (850mAh)<br>Option 3: 3.7V Single Cell Li-Poly (1000mAh) |
| Power Consumption | WISER remote with 2 K-type TC active: 1.65mA, inactive: 25µA (note: data current consumption varies with over-sampling and data packet size) @ 10 Hz |
| **Physical - Remote Unit** | |
| Dimensions | 89mm Ø x 42 mm H (standard enclosure does not include base, which is custom designed for each application) |
| Weight | 350 g |
| **Communications** | |
| Radio Frequency Transceiver Carrier | 2.45 GHz direct sequence spread spectrum, license free worldwide (2.405 to 2.480 GHz) - 16 channels, radiated power @ 3.5dBm (2.2mW) |
| RF Data Packet Standard | IEEE 802.15.4 capable, open communication architecture |
| Range for RF Link | 30m (100ft) line-of-sight |
| Base to Host Transfer | COMM Port via USB – up to 230400 baud; 8 data bits; no parity; 1 stop bit – open |
| Base Unit | USB (mounts as COM port), Dual 0-5V Analog Output Channels |
| **Environmental** | |
| Remote Operating Range | -25˚C to +125ºC (Also reference temperature measurement range) |
| TC Measurement Range | Digital: -40˚C to +1,000ºC, Analog: -40˚C to +984ºC, |
| **Electrical** | |
| Sensor Input | 2 K-type thermocouples |
| Measurement Resolution | ±1ºC |
| Data Rate | User selectable: 1Hz - 10Hz |
| Compatible Software | WISER Data Viewer or User Supplied (WDV Requires Windows 2000/XP/Vista/ Windows 7 or newer; .NET4.0; 512MB memory; 1Ghz processor or faster) |

Note: All specifications are for 10Hz data rate; 2 K-type TC's; 3.7V Li-Poly battery




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